

A Brief Introduction to Point Distortion Model

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The point distortion model is a projective model for those points in the 3-D scene would be projected on a 2-D image plane through the distorted imaging system, instead of a pinhole, through which a non-distorted image would be acquired. Relation between the non-distorted image and the distorted image can be described by the following formula, such that,

$$[x_d, y_d] = (1 + k_1(x^2 + y^2) + k_2(x^2 + y^2)^2)[x, y] \quad (1)$$

where $[x_d, y_d]$ is the distorted point coordinate, and $[x, y]$ is the non-distorted one.

The program to calculate the distorted image can be broken into 3 parts. The first part is to generate the distortion parameter $(1 + k_1(x^2 + y^2) + k_2(x^2 + y^2)^2)$, which is given by the

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1 k_matrix = 1+k(1)*(x_o.^2+y_o.^2)+k(2)*(x_o.^2+y_o.^2).^2;
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We can calculate the corresponding non-distorted coordinate $[x_d, y_d]$ of integral coordinate $[m, n]$ in distorted image after we have calculated the distorted point coordinate. It is noticed that $[x_d, y_d]$ may have decimal parts which would not fit in one certain integral point. So we add a interpolation based on the area divided by the distorted coordinate into the integral point, as is shown as follows,

(i,j+1)		(i+1,j+1)	
S_2		S_I	
(x _d ,y _d)		S_4	
S_3		S_4	
(i,j)		(i+1,j)	

These areas can be calculated using the following codes,

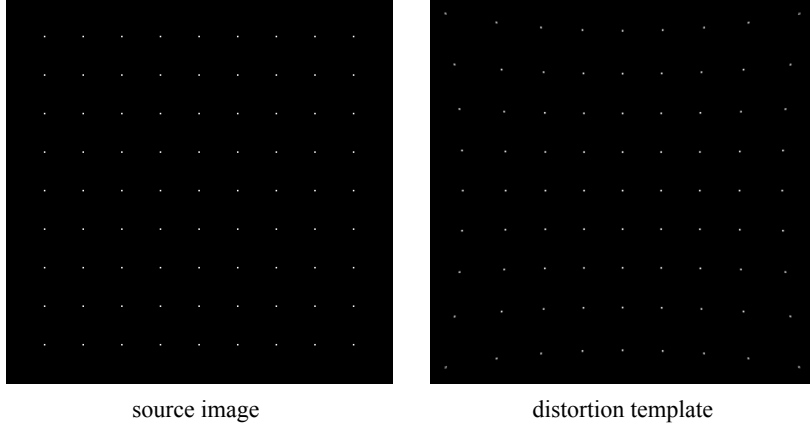
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1 s1 = (x_z+1-x_d).*(y_z+1-y_d);
2 s2 = (x_d-x_z).*(y_z+1-y_d);
3 s3 = (x_z+1-x_d).*(y_d-y_z);
4 s4 = (x_d-x_z).*(y_d-y_z);

```

where $[x_z, y_z] = [i, j]$, and $I_d(m, n) = S_1 I(i, j) + S_2 I(i + 1, j) + S_3 I(i, j + 1) + S_4 I(i + 1, j + 1)$. The distorted image $I_d(m, n)$ can be then calculated, and the reconstruction method can be applied as well.

Based on the method of generating a distorted image, we can directly search for the parameter space for an optimal polynomial parameter. The input source image and the input distortion template image is shown as follows.



The reconstruction method is to minimize the value function defined by $V(k_1, k_2) = \sum_{m,n} (I(m, n) - I_r(m, n))^2$, where $I_r(m, n)$ is calculated by parameter k_1, k_2 .